

CE/CZ4171

Internet of Things:  
Communications and Networking

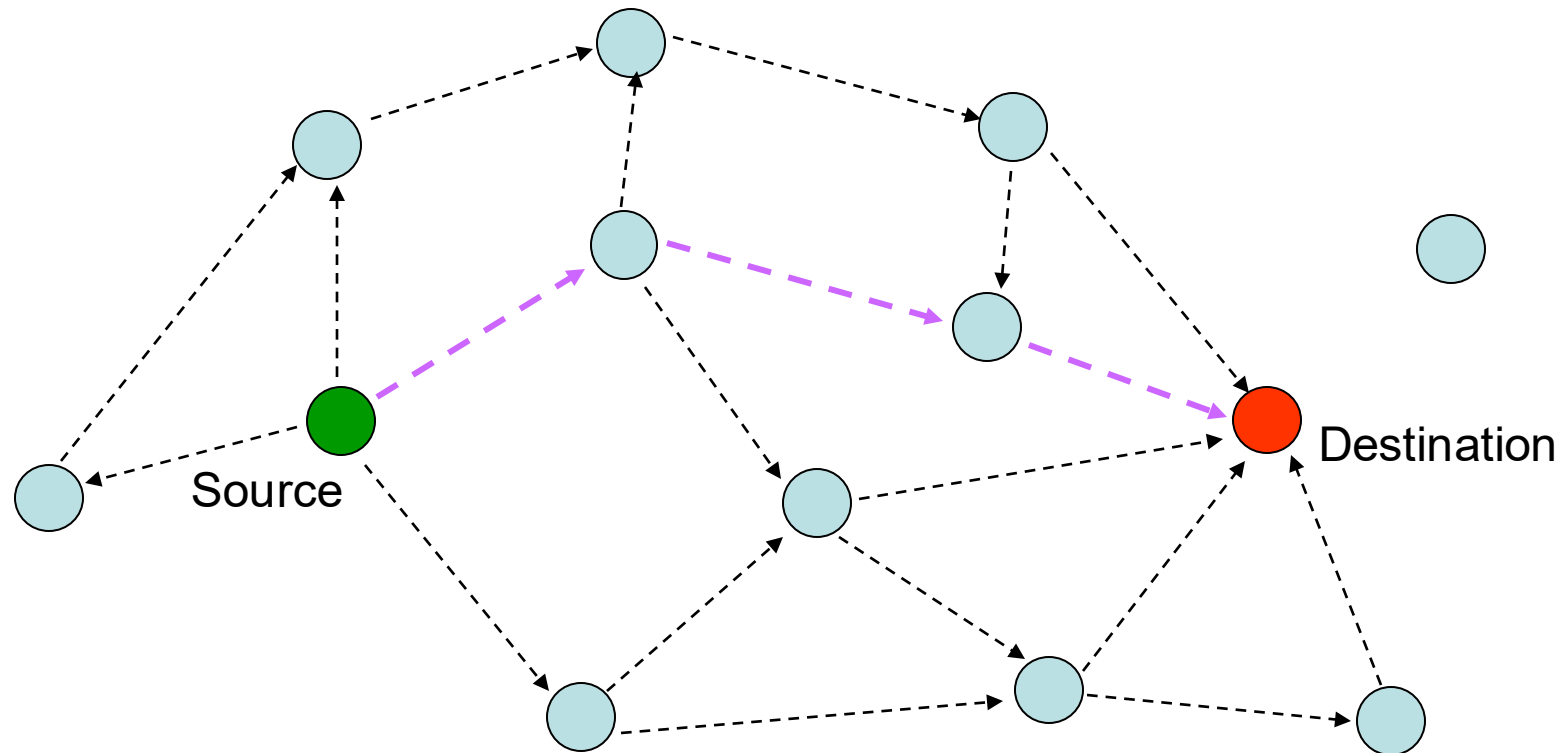
**IoT Wireless Networking**

# Outline

- **Wireless Network Routing**
- Cellular Networks

# Routing Issues

- The objective is to find an **efficient way** to find an **efficient path** from a source node to a destination node
- A non-issue in one-hop networks



# Routing Issues

- Efficient way
  - Best algorithm: fastest, smallest overhead, least energy consumption
- Efficient path
  - Best path: shortest, smallest delay, highest throughput, most reliable

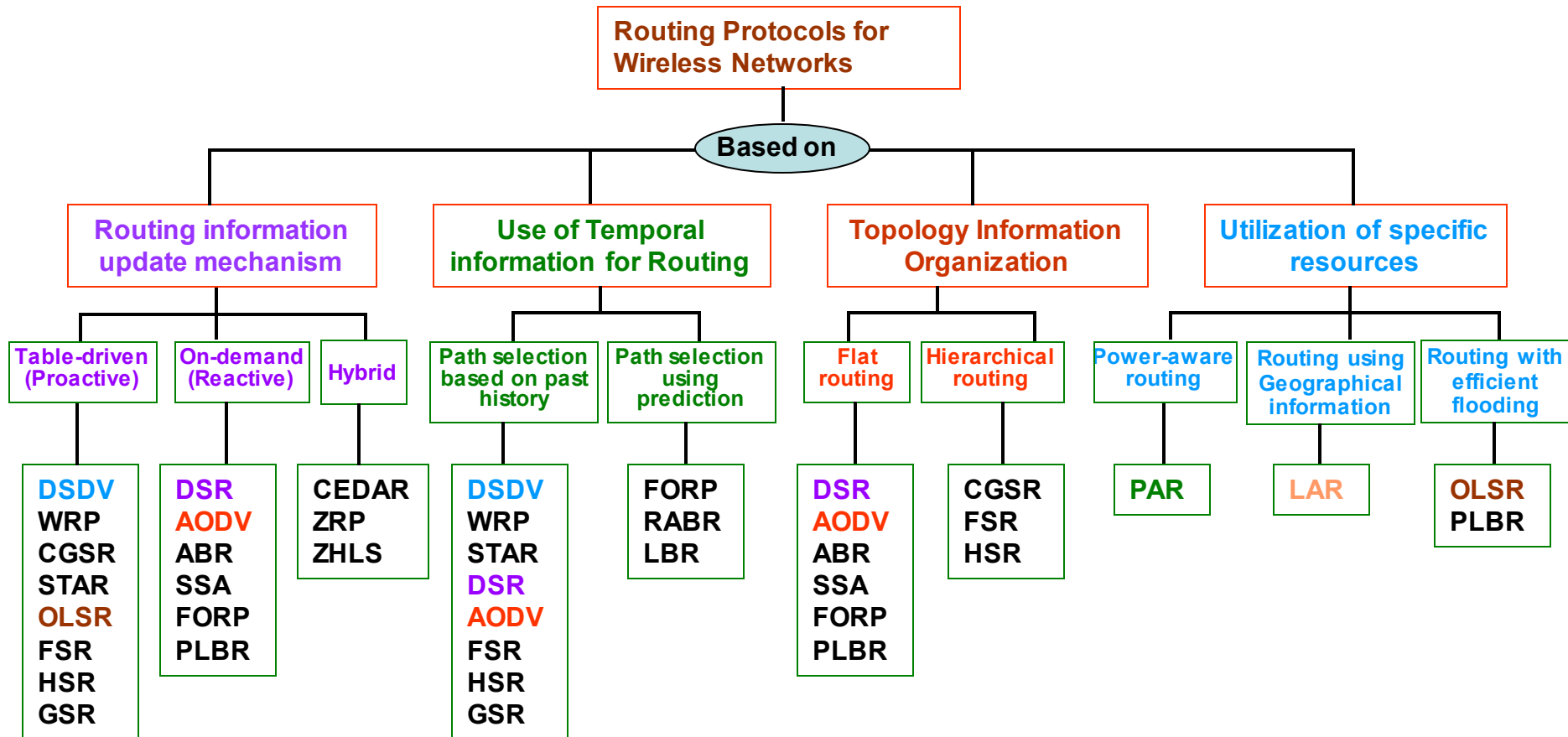
# Why different from wired routing

- **Mobility:**
  - causes frequent disruption and changes in the route
- **Absence of centralized administration:**
  - Cannot maintain and monitor network-wide information
- **Bandwidth Constraint:**
  - Overhead needs to be minimized
  - Complete topology information is costly to maintain
- **Time-varying link capacity and error rate:**
  - Needs to interact with MAC layer to find better quality links
- **Resource constraints:**
  - Limited battery and processing power → Optimal use necessary

# A well-designed routing protocol must

- Be fully distributed
- Be adaptive to frequent topology changes
- Involve minimum number of nodes for route computation and maintenance
- Have minimum connection set-up time
- Converge quickly to optimal route once topology is stable
- Be localized as global state maintenance involves huge control overhead propagation
- Be loop-free
- Limit broadcasts to minimize collisions
- Use scarce resources optimally
- Maintain only local topology

# Classification of Routing Protocols



# Routing Information Update Mechanism

- **Proactive or Table-driven:**
  - Topology information maintained by periodic exchange of routing information
  - Path finding is based on processing of already available information
- **Reactive or on-demand:**
  - No topology information maintained in advance
  - Find path as and when necessary
- **Hybrid:**
  - Combines best features of both
  - Uses topology information for nearby nodes (routing zone), for remote nodes use on-demand approach



# Use of Temporal Information for Routing

- Routing Protocol Using Past Information
  - Decision is based on information that is currently available on existence/condition/distance of links
  - Can be stable for relatively static networks
  - Overhead to maintain and reestablish link increases with mobility and topological changes
- Routing Protocol Using Future Information
  - Decision is based on expected future status of the wireless links
  - Use prediction of battery life, future location, link availability etc.

# Routing Topology

- Flat Topology
  - Uses flat addressing scheme similar to Ethernet
  - Assumes the presence of globally unique addressing mechanism in nodes
  - Is feasible for small networks
- Hierarchical Topology
  - Uses a logical hierarchy among the nodes for addressing
  - Can be based on geographical information on hop distance
  - Scalable and suitable for large networks

# Utilization of Specific Resources

- Power Aware routing
  - Routing decision based on minimizing battery consumption; either locally, or globally
- Geographical Information Assisted routing
  - Effectively utilize location information of nodes
  - Reduces control overhead

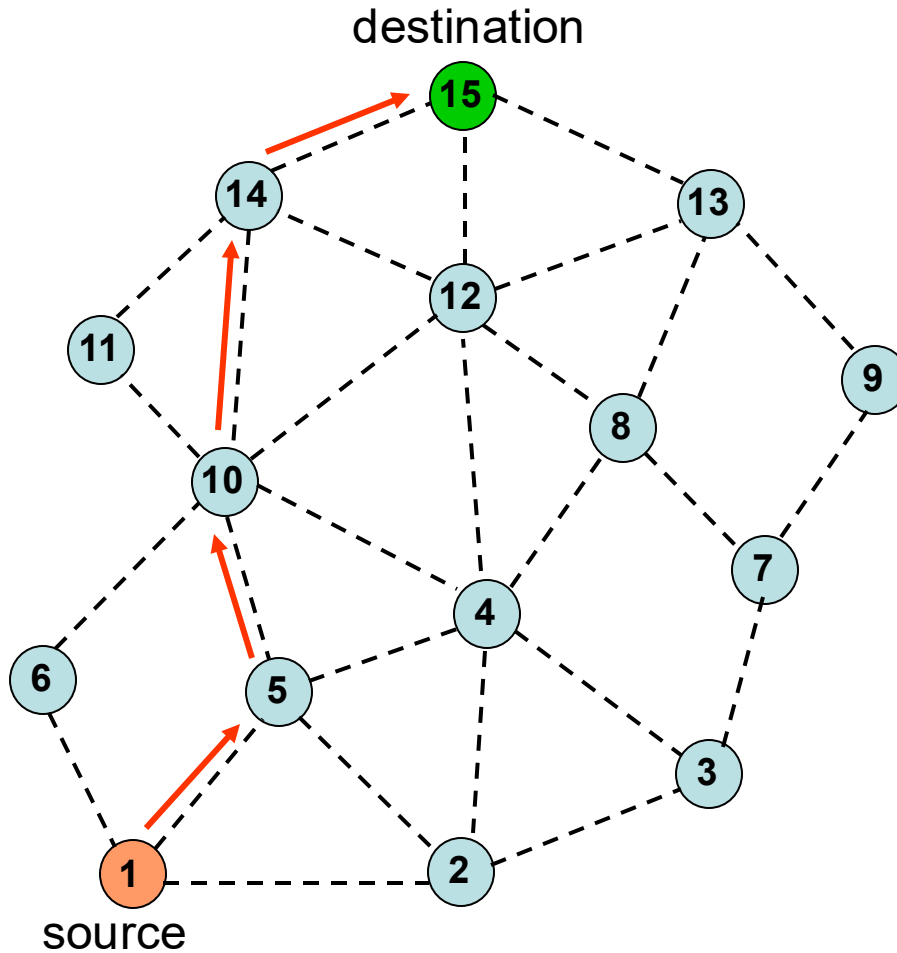
# Destination Sequenced Distance Vector Routing (DSDV)

- Based on Distributed Bellman-Ford algorithm
- Table driven
- Table contains shortest distance and the first node on the shortest path to every other node in the network
- Update of each entry in the routing table is tagged with an increasing sequence number which prevents routing loops
- Tables are exchanged between neighbors at regular intervals
- In addition, Tables are forwarded if a node observes significant change in local topology
- Table updates can be incremental updates or full dumps

# Destination Sequenced Distance Vector Routing (DSDV)

- The data broadcast by each mobile host includes:
  - the destination's address
  - its new sequence number (itself as destination)
  - the number of hops required to reach the destination
  - the sequence number of the information received regarding the destination
- Updates are initiated by a destination with a new sequence number
- Upon receiving updated table, a node updates its table if the sequence no. is higher than current value

# Example of DSDV protocol



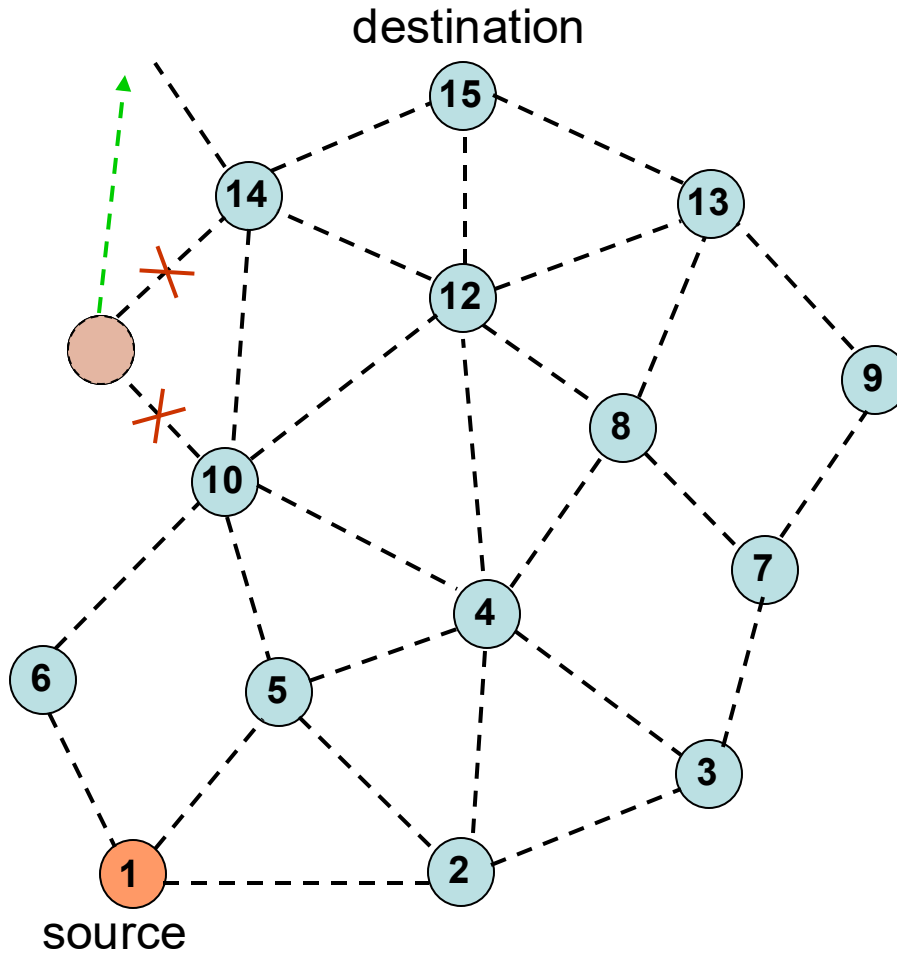
| Destn | Next node | Dist | Seq no. |
|-------|-----------|------|---------|
| 2     | 2         | 1    | 22      |
| 3     | 2         | 2    | 26      |
| 4     | 5         | 2    | 32      |
| 5     | 5         | 1    | 134     |
| 6     | 6         | 1    | 144     |
| 7     | 2         | 3    | 162     |
| 8     | 5         | 3    | 170     |
| 9     | 2         | 4    | 186     |
| 10    | 6         | 2    | 142     |
| 11    | 6         | 3    | 176     |
| 12    | 5         | 3    | 190     |
| 13    | 5         | 4    | 198     |
| 14    | 6         | 3    | 214     |
| 15    | 5         | 4    | 256     |

Routing table of node 1

# Handling Link Breaks

- A link can break because of
  - Mobility
  - Node shutdown / failure
  - Poor channel condition
- A node detecting broken link initiates table update as follows
  - The broken link is assigned a weight of infinity ( $\infty$ )
  - Broadcasts a table update message with a higher sequence no. than its current sequence no.
  - For a link break update, an odd sequence number is used to differentiate it from normal updates
  - Each node forwards the received message to its neighbors

# Link Break Update Example



| Destn | Next node | Dist | Seq no. |
|-------|-----------|------|---------|
| 2     | 2         | 1    | 22      |
| 3     | 2         | 2    | 26      |
| 4     | 5         | 2    | 32      |
| 5     | 5         | 1    | 134     |
| 6     | 6         | 1    | 144     |
| 7     | 2         | 3    | 162     |
| 8     | 5         | 3    | 170     |
| 9     | 2         | 4    | 186     |
| 10    | 6         | 2    | 142     |
| 11    | 5         | 4    | 180     |
| 12    | 5         | 3    | 190     |
| 13    | 5         | 4    | 198     |
| 14    | 6         | 3    | 214     |
| 15    | 5         | 4    | 256     |



# Link Break Update Example

- Node 11 moves from its current position
- Node 10 realizes this event after a while (e.g. no acks from 11)
- Node 10 sets all path lengths passing through node 11 as  $\infty$ , and broadcasts it to neighbors
- Node 1 also sets its distance to node 11 as  $\infty$
- Node 11 broadcasts update message
- Node 14 receives this message, and propagates it
- All nodes receiving the update message originated by node 11 with higher sequence number, updates the distance to node 11

# Advantages and Disadvantages of DSDV

- Advantages
  - As routing information is readily available in all nodes, routing delay is minimized
  - Existing wired network protocols use similar routing tables – can be adapted easily
- Disadvantages
  - Updates due to broken links lead to heavy control overhead – can choke network if mobility is high
  - Control overhead is proportional to the number of nodes – hence grows with network size → not scalable
  - Updates are initiated by destination nodes → can cause delay and stale routing information at nodes

# Dynamic Source Routing (DSR) Protocol

- On-demand routing protocol
- Designed to restrict bandwidth consumption by periodic table-updates
- It is beacon-less –i.e., does not send periodic ‘Hello’ packets to neighbors to inform of its presence
- When a source node has data packet to send, it initiates a *RouteRequest* (RREQ) Packet with a sequence number
- This packet is flooded through the network by repeated broadcasting by other nodes
- A node rebroadcasts a RouteRequest packet if
  - It has not already forwarded the same packet (checks the seq. no.)
  - It is not the destination
  - The Time-to-live (TTL) counter of the packet has not exceeded
- The forwarded RouteRequest packet contains the path it has already traversed
- The sequence number check avoids loops and multiple transmissions of the same packet received through multiple paths

# DSR Protocol

- A destination Node after receiving the first RouteRequest packet, initiates a *RouteReply* (**RREP**) packet
- The **RREP** packet is sent to the source node through the reverse path
- The route is constructed as the **RREP** message retraces the path to the source
- Source now includes this route information in the data packet sent to the destination

# DSR Optimizations

- Using Route Cache

- Intermediate nodes store in **route cache** all possible information extracted from the source route contained in a data packet
- When a node receives a new **RREQ** message, it checks its route cache. If it has route to the destination, then it sends the **RREP** message with the complete path information.
- Nodes also update the route cache when they come to know of broken routes (by promiscuous listening)

- Exponential Backoff

- In case of a network partitioning, many nodes may initiate RouteRequest
- Exponential backoff is used to avoid frequent flooding in such cases

# DSR optimizations (cont.)

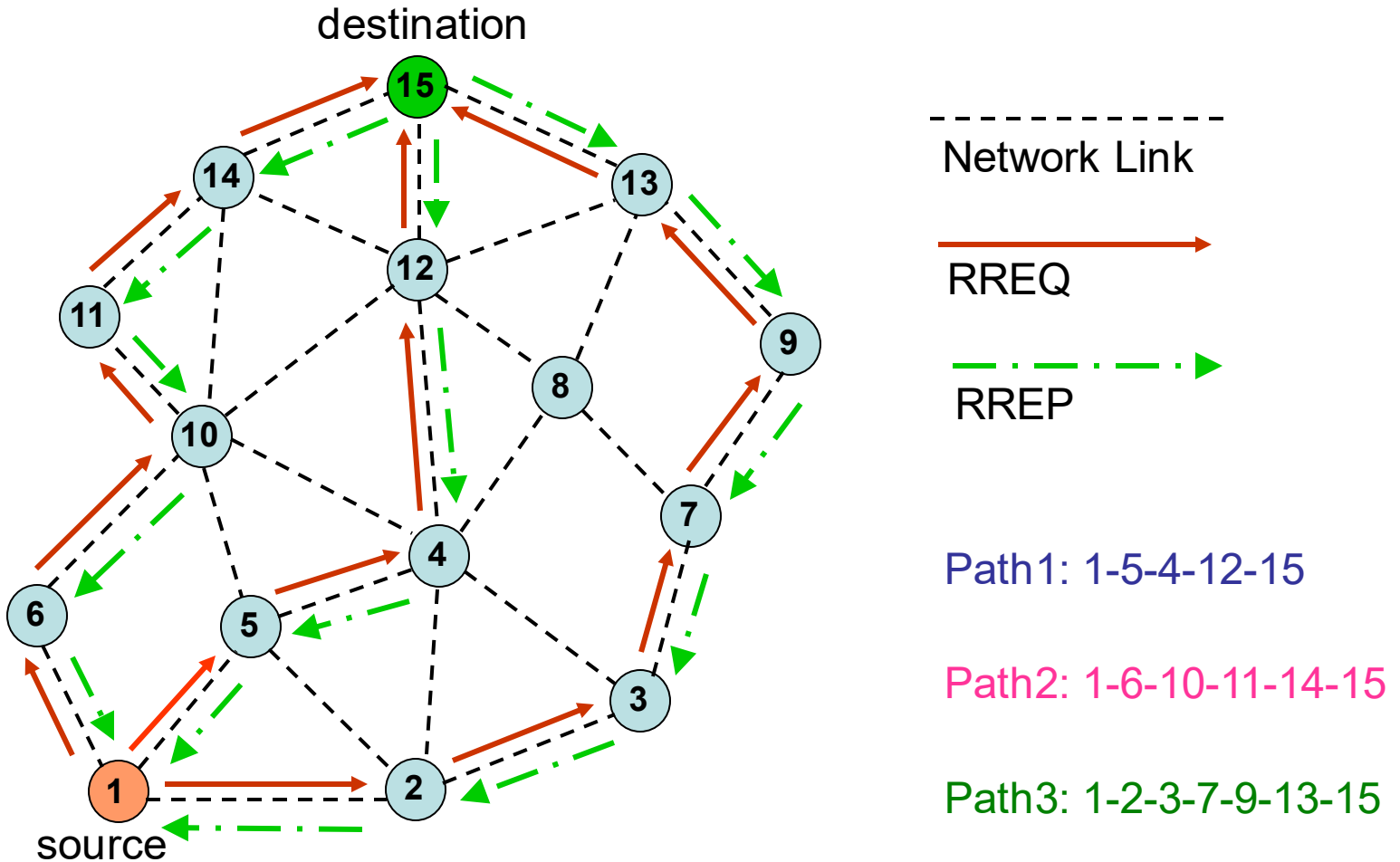
- Optimal selection of route

- Since an intermediate node can send a **RREP** based on its route cache, a source node may receive multiple route replies.
- In this case, the source chooses the latest and best route

- Handling Broken Links

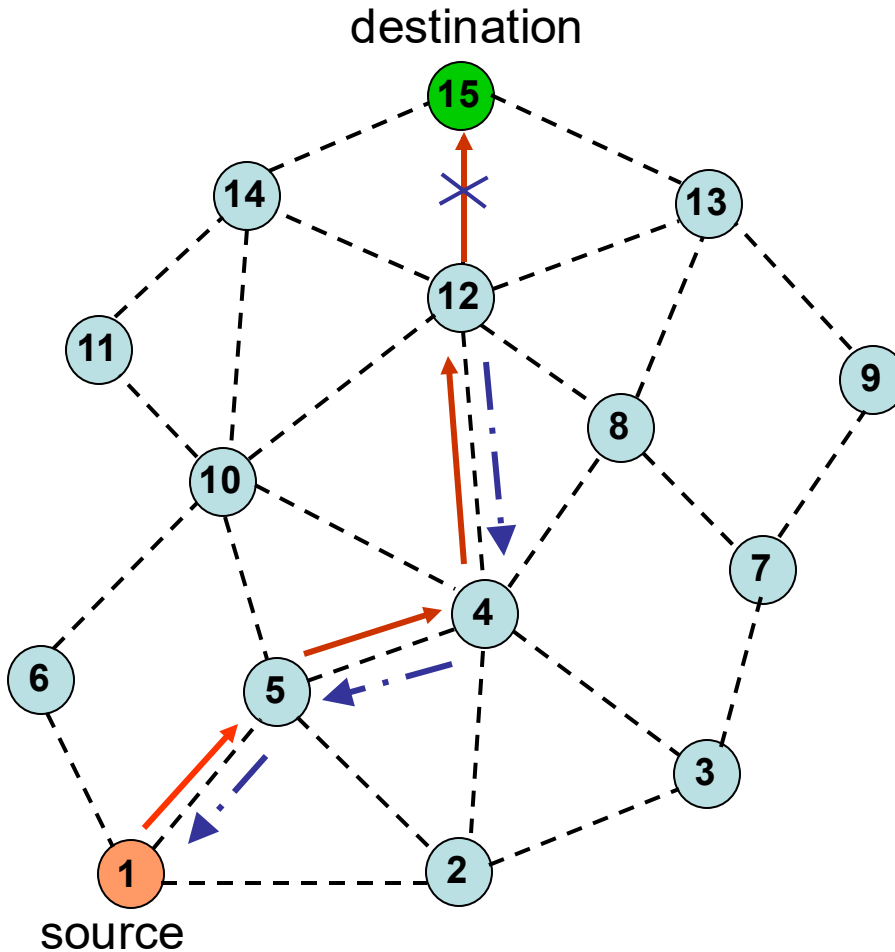
- If a link is broken, a **RouteError (RERR)** message is generated from the node adjacent to the broken link to inform the source node
- The source reinitiates route establishment procedure
- The affected cached entries in nodes are removed when a **RERR** message is received

# Example of DSR protocol



Topology graph of the network

# Broken Link in DSR Example



- Node 12 detects broken link
- Node 12 generates RERR message
- Source Node reinitiates route establishment procedure.



# Advantages and Disadvantages of DSR

- **Advantages**

- Eliminates periodic routing update message → bandwidth efficient
- Route cache mechanism also reduces control overhead and makes route establishment faster

- **Disadvantages**

- Route maintenance mechanism does not locally repair a broken link
- Stale route cache information can result in inconsistencies
- Connection setup delay is higher than table-driven approach
- Performance degrades rapidly with increasing mobility
- Data Packet carries complete path – this routing overhead is proportional to path length

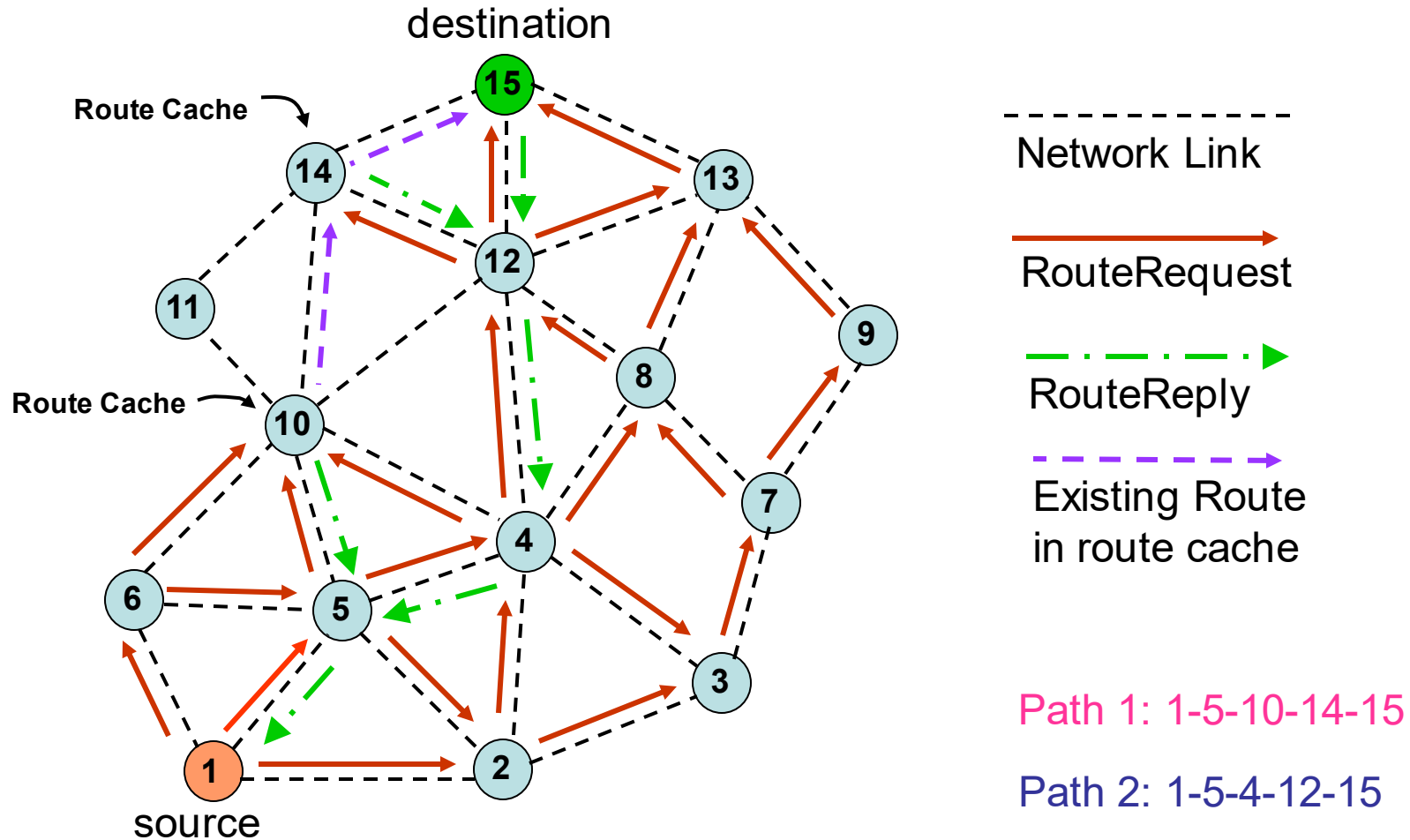
# Ad Hoc On-Demand Distance vector Routing (AODV)

- Routing initiated only when required
- Like DSDV, uses destination sequence number to identify most recent path
- Unlike DSR, data packet does not carry complete path information
- Nodes send periodical **beacons** or '**Hello**' message to neighbors
- Source node and intermediate nodes store next-hop information corresponding to each flow
- It uses flooding technique similar to DSR to propagate **RREQ**
- It is possible to receive multiple routes for each request

# Ad Hoc On-Demand Distance vector Routing (AODV)

- **RREQ** packet contains the following
  - Source ID (SrcID)
  - Destination ID (DestID)
  - Destination Sequence number (DestSeqNum)
  - Broadcast ID (BcastID)
- An intermediate node, checks for duplicates. If new request, then
  - Records the previous node address and BcastID, and
  - Either Forwards it, or
  - Sends back **RREP** if it has valid route to destination
- The saved info. is deleted if **RREP** not received before a timer expires
- Information about previous node is also saved when forwarding a **RREP**

# AODV Example



# AODV Example (cont.)

- Node 1 initiates **RREQ** with
  - SrcID = 1, DestID = 15, DestSeqNo = 3
- **RREQ** is forwarded by nodes 2, 5, 6 to nodes 3, 4, 10
- Nodes 3 and 10 already have routes to node 15
- DestSeqNo at node 10 = 4, at node 3 = 1. So, only node 10 replies with **RREP** message
- Node 3 forwards **RREQ** to node 7
- **RREQ** reaches node 15 through 4-12-15, and it sends a **RREP**
- Node 14 on receiving RREQ from node 12, sends **RREP** to node 12
- Node 12 sees two RREPs, and forwards the one from node 15 as it has higher sequence no.

# Advantages and Disadvantages of AODV

- Advantages
  - On-demand reduces routing overhead
  - DesttSeqNo. Helps maintain latest route
- Disadvantages
  - Multiple RREP can lead to high control overhead
  - Periodic beaconing leads to bandwidth consumption

# Location-Aided Routing (LAR)

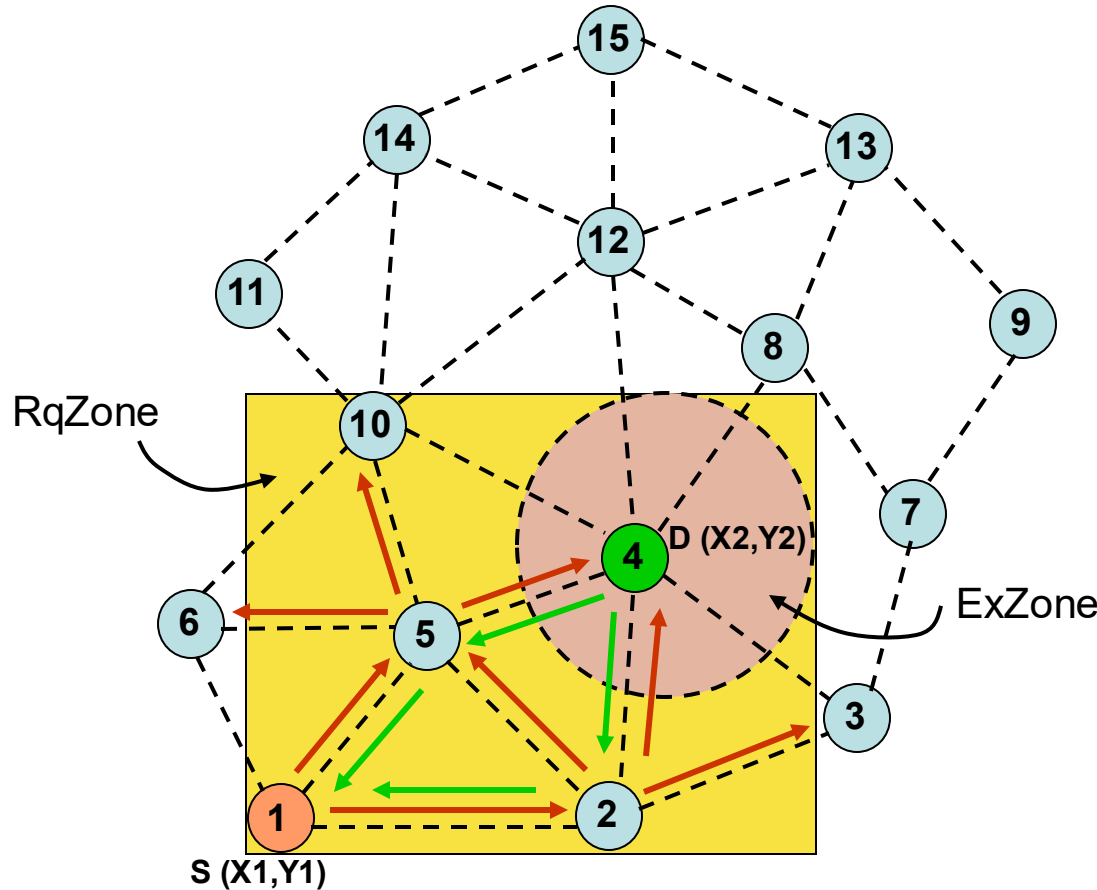
- Uses location information to reduce control overhead by Selective forwarding
- Assumes availability of Global Positioning System (GPS)
- Designates two geographic regions:
  - **RequestZone (RqZone)**: The region within which the routing request should be restricted
  - **ExpectedZone (ExZone)**: The region where the destination node is expected, based on its location and mobility information
- In the beginning, ExpectedZone spans entire network (as no information is available)
- RqZone can be extended if the routing attempt fails within a timeout

# Location-Aided Routing (LAR)

- Explicitly specifies RqZone in the RREQ packet
- RqZone is the smallest rectangle that encloses
  - both the source (S) and the ExZone when S is outside ExZone
  - The ExZone when S is within ExZone
- Every intermediate node forwards RREQ only if it is within RqZone



# Location-Aided Routing (LAR) Example



- Source S broadcasts RREQ to nodes 2, 5, 6
- 2, 5 in RqZone, so they forwards RREQ
- 6 discards RREQ as it is outside RqZone
- Destination D (node 4) generates RREP which includes:
  - Current time
  - Current location of D
  - Current speed of D

# Advantage and Disadvantage of LAR

- Advantage:
  - Control overhead substantially reduced by limiting flooding of RREQ using geographical information
- Disadvantage:
  - Depends on availability of GPS

# Cluster-Head Gateway Switch Routing Protocol (CGSR)

- Uses hierarchical network topology
- Table-driven
- Network organized into **clusters**
- A special node called **cluster-head** performs coordination among member nodes
- All communication between members of a cluster goes through the cluster-head
- Communication between clusters is performed through **Gateways**, that are common nodes i.e. member of two (or more) clusters.

# Election of Cluster-Head

- Dynamically elected using **Least Cluster Change (LCC) Algorithm**

## LCC Algorithm:

1. Initially, every node is a cluster-head
2. If two cluster-heads are in transmission range of each other, then choose the one with the highest connectivity (or lowest ID) as the new cluster-head

**All members are within one-hop distance from cluster-head**

# Bandwidth Use among Clusters

## Inter-cluster

- Clustering facilitates bandwidth reuse
- Each cluster can operate using a different spreading code on a CDMA system → gateway nodes must be able to use multiple codes

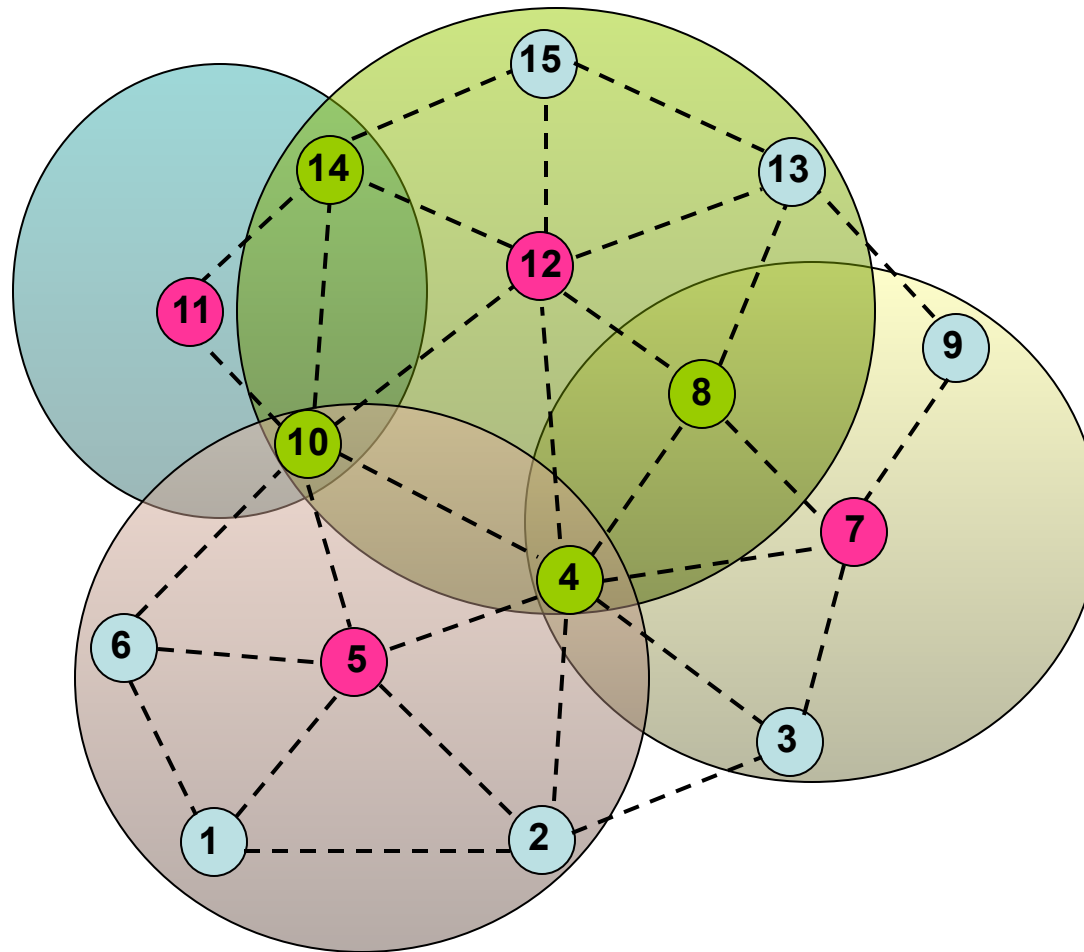
## Intra-cluster

- Cluster-head coordinates channel access based on a token-based polling protocol
- A **Gateway conflict** occurs when a token is passed to gateway over a spreading code while it is tuned to another code
- Gateway conflict can be avoided by simultaneous tuning to multiple codes

# Routing in CGSR

- An extension of DSDV
- Every member-node maintains a cluster-member table containing the destination cluster-head for every node in the network
- The cluster-heads maintain a routing table that contains the gateway node for every destination cluster
- When a node gets a token from the cluster-head it forwards the packet to the cluster-head with the destination cluster-head information.
- The cluster-head looks up its routing table to identify the gateway node to which the packet should be forwarded

# CGSR: Example



Path from node 2 to  
node 13:

2-5-4-12-13



Cluster-head



Gateway



Member-node

# Advantages and Disadvantages of CGSR

- Advantages:
  - Better bandwidth utilization because of partial coordination among nodes
  - Possible to implement priority scheduling through token scheduling
- Disadvantages
  - Possibly increased path length
  - Instability at high mobility because of frequent change in cluster-head
  - More resources (multiple code receiver) necessary at gateway nodes to avoid gateway conflict
  - Cluster-heads are loaded more than member nodes → can cause fast battery drain



# Summary

- Characteristics of ideal routing Protocol
- Classification based on various criteria
- Protocols studied:
  - DSDV
  - DSR
  - AODV
  - LAR
  - OLSR
  - CGSR

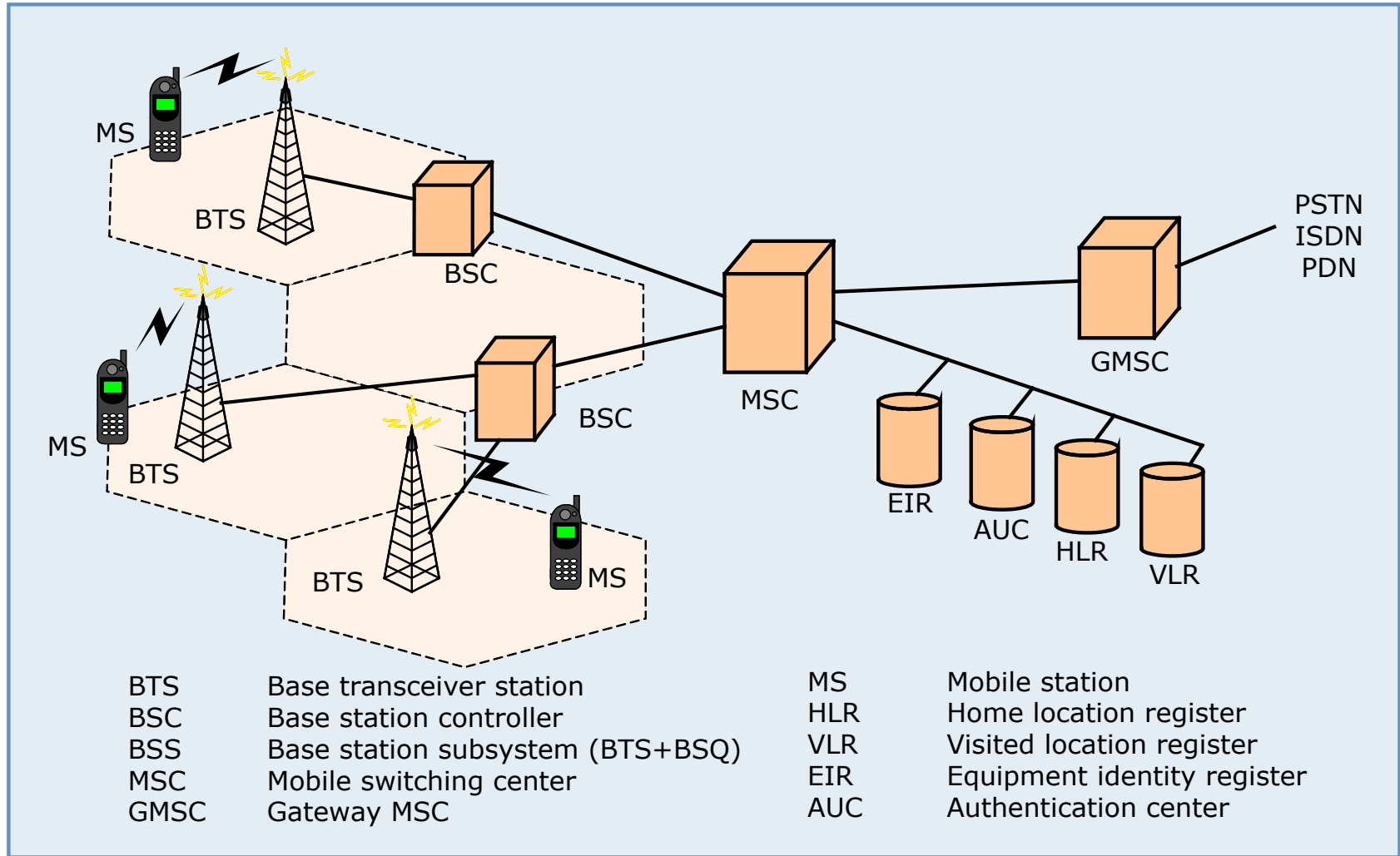
# Outline

- Wireless Network Routing
- **Cellular Networks**

# Cellular Networks

- Cellular System Architecture
- Frequency Reuse Concept
- Hexagon Geometry
- Co-Channel Interference
- Capacity Expansion Techniques

# The Architecture of a Cellular Network



# Frequency Reuse Concept

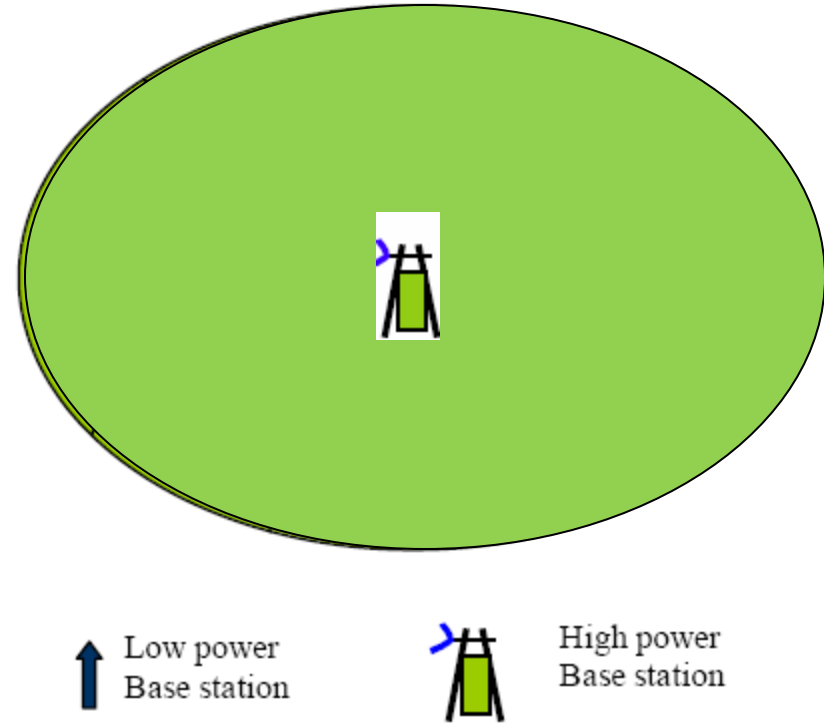
- One of the major problems in wireless networks is to support a large number of users with a limited frequency spectrum.
- In the 1970s, the Bell mobile system in New York could only support 12 simultaneous calls over a thousand square miles.
- Frequency reuse concept
  - Spatially reusing the available spectrum so that the same spectrum can support >1 users separated by distance.
  - In free space, the signal strength of the electromagnetic wave attenuates as the square of the distance.

$$\text{Signal Strength} = k \times \frac{1}{(\text{distance})^2}$$

- Same frequency can be re-used at two locations sufficiently far away with acceptable interference level.

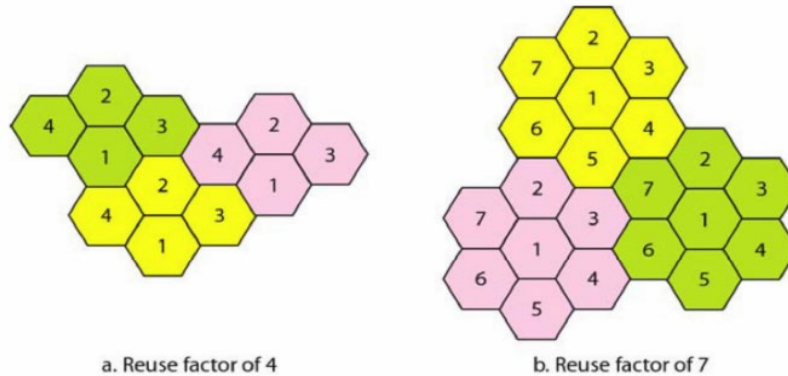
# Frequency Reuse Concept

- “Simplest” way to provide mobile communications
  - Deploy one high-power base station with sufficient power to provide coverage over an entire area.
- Cellular Concept
  - Divide the area of coverage into a number of smaller contiguous areas (Cell).
  - The MS (Mobile Station) in each area is served by a lower powered BS (Base Station).



# Frequency Reuse Concept

- Clusters
  - Contains a number of cells, called the **cluster size** or **Frequency Reuse Factor**
  - Each cluster utilizes the entire available radio spectrum
  - Same frequencies are reused in each cluster



- Co-channel interference
  - Interference due to the use of *same* frequency in cells of different clusters
  - *Co-channel Cells*: cells (from different clusters) that use the same frequency

# Frequency Reuse Concept

- By definition, the number of users per cell is

$$M = \frac{W}{W_c} \cdot \frac{1}{N}$$

- W: total bandwidth available
- $W_c$ : Carrier bandwidth (channel bandwidth)
- N: frequency reuse factor (or cluster size)



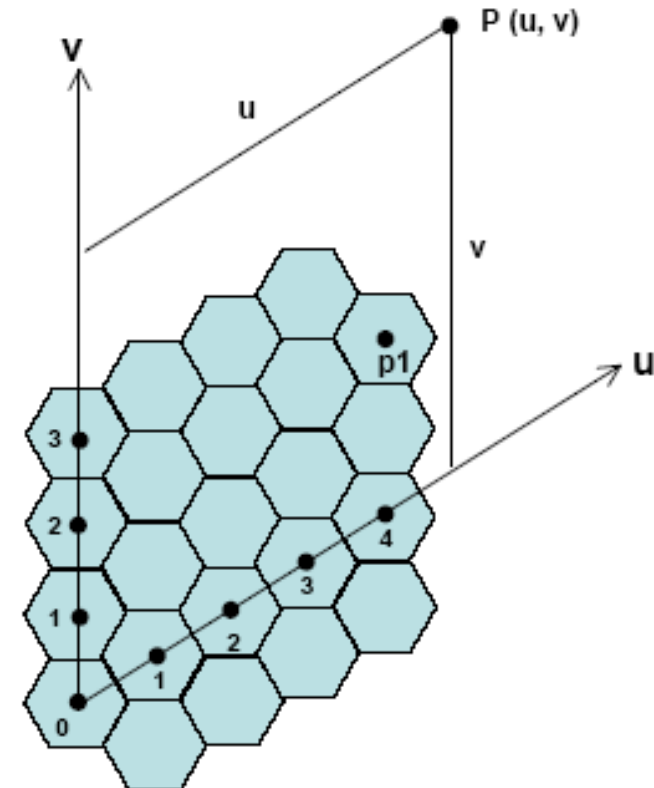
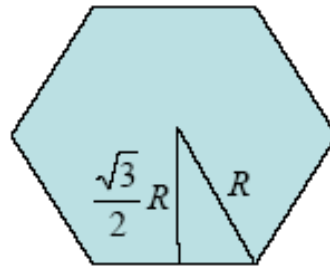
# Frequency Reuse Concept

Example: We want to provide a radio communication service to a city. The total bandwidth available is 25 MHz, and each user requires 30 KHz of bandwidth for voice communication.

- Case 1: One high power base station
  - Maximum number of simultaneous users =  $25 \text{ MHz} / 30 \text{ KHz} = 833$
- Case 2: Cellular topology: Use 20 lower power antennas, cluster of 4 cells, 5 clusters and 20 cells in total.
  - number of users per cell: 
$$M = \frac{W}{W_c} \cdot \frac{1}{N} = \frac{25 \text{ MHz}}{30 \text{ KHz}} \cdot \frac{1}{4} = 208$$
  - Total number of users =  $208 \times 4 \times 5 = 4160$
  - 5 times improvement!

# Hexagon Geometry

- Hexagon Geometry Coordinate
  - u-axis:  $30^\circ$  angle to the x-axis
  - v-axis: same as y-axis
  - Unit distance along either axis is equal to  $\sqrt{3}R$
- For all integer values of  $u, v$ , the points  $(u,v)$  are centers of hexagons.



Coordinate of point  $P_1 = (4, 2)$

# Hexagon Geometry

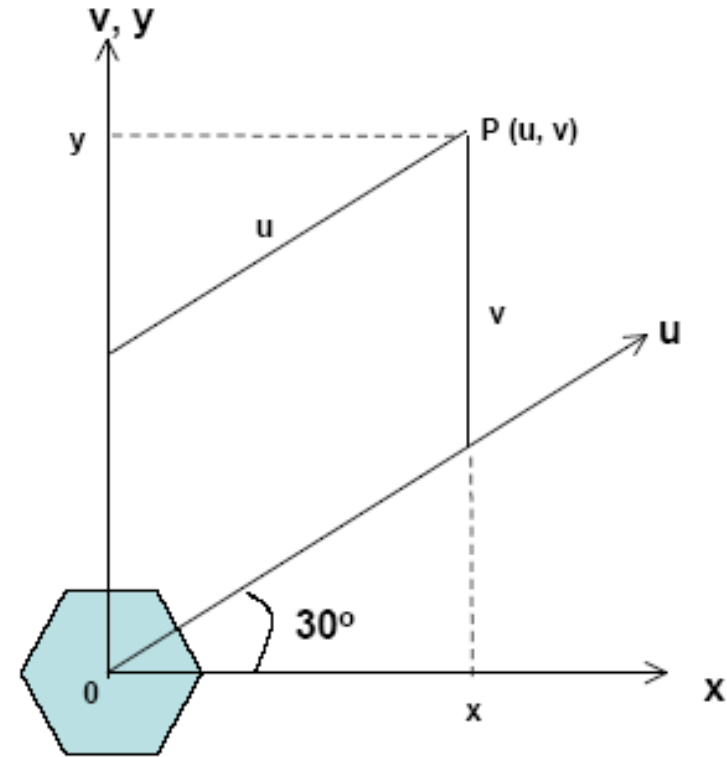
- Projection onto the Cartesian coordinates system

$$x = \sqrt{3}R \times (u \cos 30^\circ) = \sqrt{3}R \times \left(\frac{\sqrt{3}}{2}u\right)$$

$$y = \sqrt{3}R \times \left(v + u \sin 30^\circ\right) = \sqrt{3}R \times \left(v + \frac{1}{2}u\right)$$

- Distance between cell centers
  - Given two points  $(u_1, v_1)$  and  $(u_2, v_2)$ , the distance between them is

$$\begin{aligned} d_{12} &= \sqrt{(y_2 - y_1)^2 + (x_2 - x_1)^2} \\ &= \sqrt{3}R \times \sqrt{(u_2 - u_1)^2 + (u_2 - u_1)(v_2 - v_1) + (v_2 - v_1)^2} \end{aligned}$$



# Hexagon Geometry

- For  $u$  and  $v$  integers (cell centers), we define  $i$  and  $j$ , both integers such that:  $i = (u_2 - u_1)$   $j = (v_2 - v_1)$   
 $i$  and  $j$  are called shift parameters. The distance between cell centers become

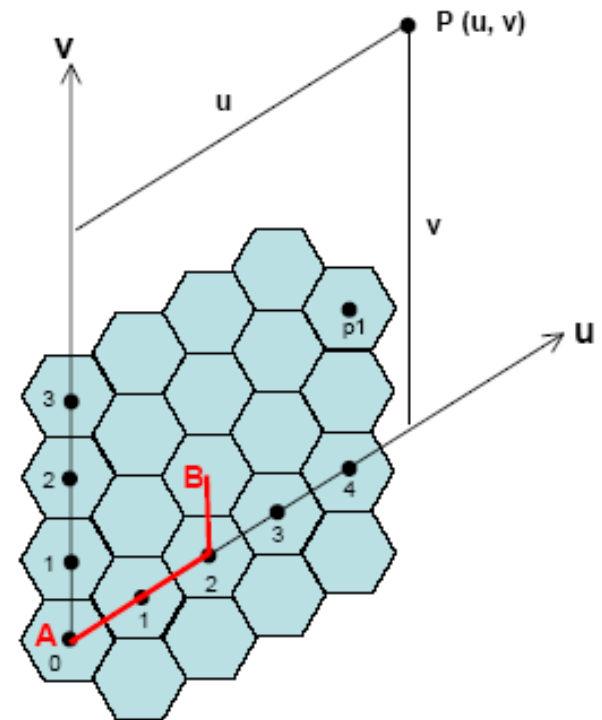
$$d = \sqrt{3}R \times \sqrt{(i^2 + i \times j + j^2)}$$

- Example: Distance between cell A and B is ?

$$i=2, j=1$$

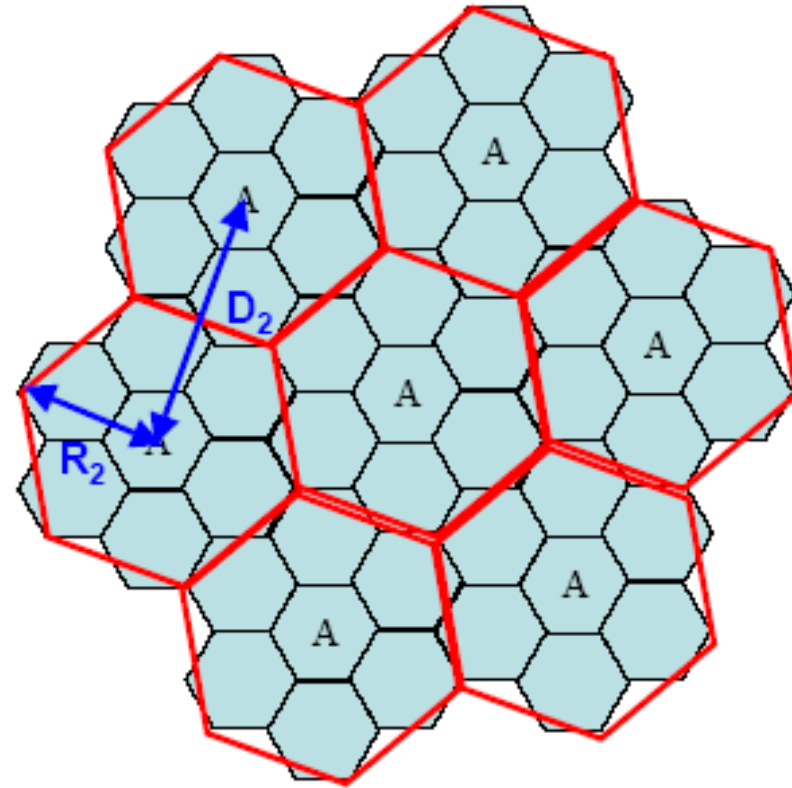
$$d = \sqrt{3}R \sqrt{2^2 + 2 \times 1 + 1^2}$$

$$d = \sqrt{21}R$$



# Hexagon Geometry

- Cluster size
  - Q: Can cluster size  $N$  be an arbitrary value?
  - An: No,  $N = i^2 + ij + j^2$
  - Approximate the cluster to a large hexagon. The large hexagons provide complete coverage over the area with no overlap and gap. The area of the large hexagon is exactly the total area of  $N$  cells.



Cluster size  $N = 7$

# Hexagon Geometry

- Let  $R_1$  and  $D_1$  denote the radius and the distance between adjacent hexagons for small hexagon. Let  $R_2$  and  $D_2$  denote the radius and the distance between adjacent hexagons for big hexagon. We have

$$D_1 = \sqrt{3}R_1 \quad D_2 = \sqrt{3}R_2$$

$$D_2 = \sqrt{3}R_1\sqrt{i^2 + ij + j^2}$$

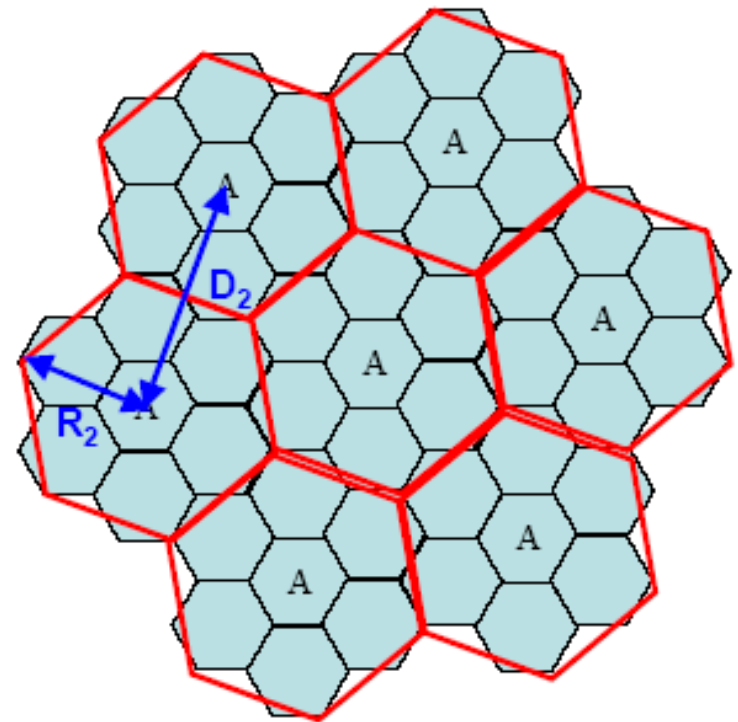
$$R_2 = R_1\sqrt{i^2 + ij + j^2}$$

$$A_2 = \frac{3\sqrt{3}}{2}R_2^2 \quad A_2 \propto R_2^2 \quad A_1 \propto R_1^2$$

$$A_2 = NA_1 \quad R_2^2 = NR_1^2$$

$$R_1^2 = i^2 + ij + j^2 = NR_1^2$$

$$N = i^2 + ij + j^2$$



Cluster size  $N = 7$

# Hexagon Geometry

- Co-channel reuse ratio  $q$ :

$$q = \frac{D}{R} = \sqrt{3N}$$

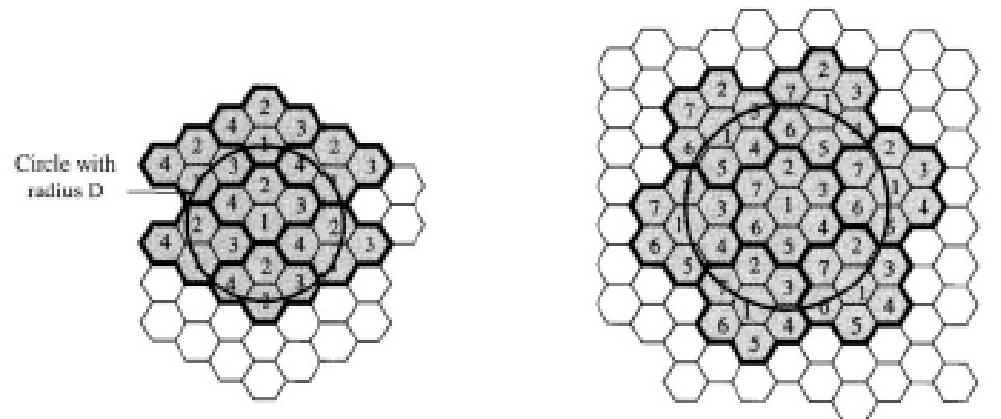
$$D_2 = \sqrt{3}R_1\sqrt{i^2 + ij + j^2} = \sqrt{3}R_1\sqrt{N}$$

where:

- $D$ : Distance between centers of closest co-channel cells
- $R$ : Cell radius
- $N$ : cluster size (in the form of  $N = i^2 + ij + j^2$ , where  $i, j$  are integers) ( $N=1,3,4,7,9,12,13,16,19, \dots$ ).

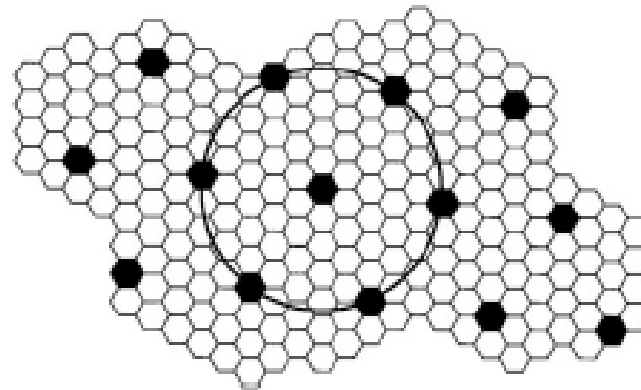
# Hexagon Geometry

- There are always 6 co-channel cells at the same distance.



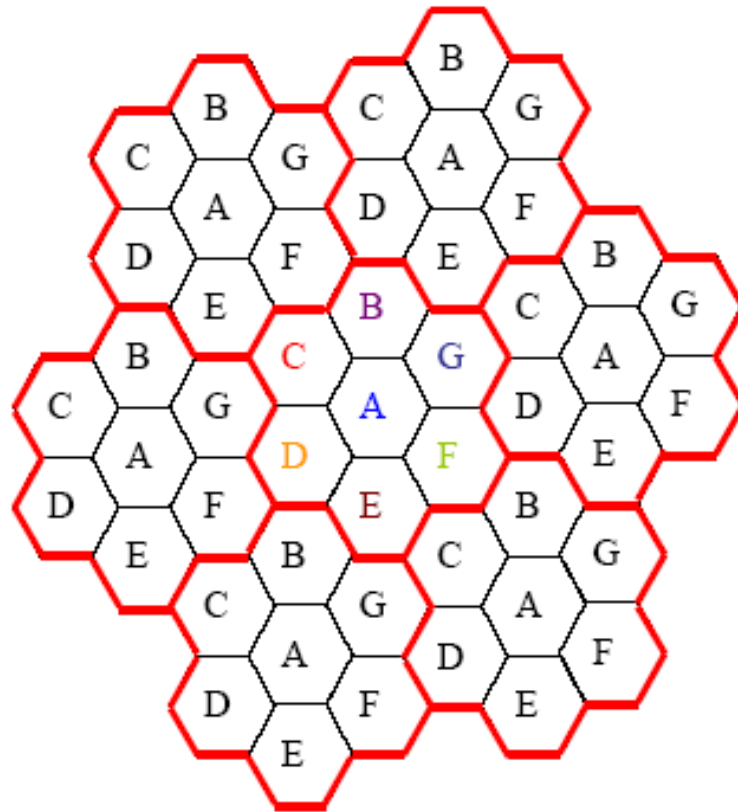
(a) Frequency reuse pattern for  $N = 4$

(b) Frequency reuse pattern for  $N = 7$





# Co-Channel Interference



# Co-Channel Interference

- Signal-to-interference ratio (SIR) is measured by

$$\frac{S}{I} = \frac{S}{N_0 + \sum_{i=1}^m I_i}$$

$S$  : the desired signal power

$N_0$  : the noise (negligible)

$m$  : the number of interferers

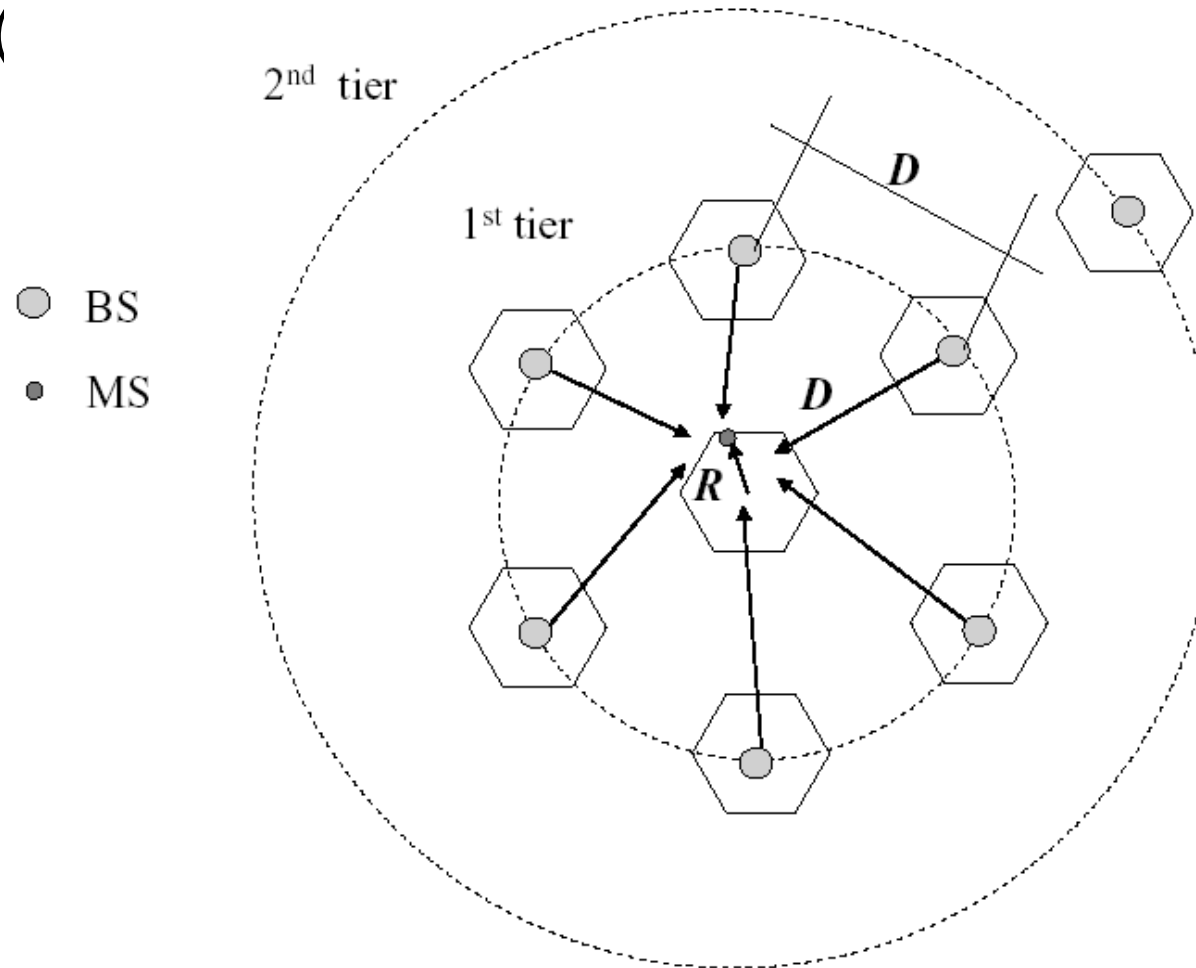
$I_i$  : the power of the  $i$ -th interferer.

- Considering downlink co-channel interference, assuming signal loss is only due to path loss and all the BSs have the same transmission power, then  $S \propto PD_0^{-n}$ ;  $I_i \propto PD_i^{-n}$  . Hence, SIR becomes

$$\frac{S}{I} = \frac{D_0^{-n}}{\sum_{i=1}^m D_i^{-n}}$$

# Co-Channel Interference

- Co-channel interference receiving at the mobile



# Co-Channel Interference

- Simple Analysis

- For any  $N$ , the number of co-channel interferers in the 1st tier is 6 and that in the 2nd tier is also 6. Normally, we can neglect the interference from the 2nd tier and above because they are far away.
- Assuming the distance between MS and the desired BS is cell radius,  $R$ , and the distance between MS and any of the interfering BSs is  $D$ . SIR becomes

$$\frac{S}{I} = \frac{D_0^{-n}}{\sum_{i=1}^m D_i^{-n}} = \frac{D^n}{m \cdot R^n} = \frac{D^n}{6R^n} = \frac{q^n}{6} = \frac{(\sqrt{3N})^n}{6}$$

- If  $n=4$ ,  $SIR = 3N^2 / 2$ . In terms of dB,  $SIR = 10 \log_{10}(3N^2 / 2) = 1.76 + 20 \log N$

# Co-Channel Interference

- Example: If a SIR of 10 dB is required for a cellular system, what is the optimum cluster size, if the path loss exponent  $n=3$ ? Using simple analysis assumptions.

For  $N = 1$ ,  $SIR = -0.6$  dB

For  $N = 3$ ,  $SIR = 6.53$  dB

For  $N = 4$ ,  $SIR = 8.41$  dB

For  $N = 7$ ,  $SIR = 12.1$  dB

For  $N = 9$ ,  $SIR = 13.69$  dB

For  $N = 12$ ,  $SIR = 15.56$  dB

For  $N = 13$ ,  $SIR = 16.08$  dB

Solution: The optimal cluster size is 7. Why?

# Co-Channel Interference

- Discussion on Frequency Reuse Factor N
  - From the SIR viewpoint, we want to maximize N in order to maximize the SIR.

$$SIR = (\sqrt{3N})^n / 6$$

- From the capacity viewpoint, we want to minimize N in order to maximize the number of channels within a cell.

$$M = \frac{W}{W_c} \cdot \frac{1}{N}$$

- Tradeoff between system capacity and performance.
    - Smaller N -> bigger capacity but higher interference.
    - Bigger N -> smaller capacity but lower interference.

# Cellular Networks

- Cellular System Architecture
- Frequency Reuse Concept
- Hexagon Geometry
- Co-Channel Interference